INCH-POUND

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SUPERSEDING
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w/AMENDMENT 1
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DETAIL SPECIFICATION

# CHARGE, DEMOLITION: SHAPED, 15 LB, M2A4
# PARTS, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the parts, loading, assembling and packing of the Charge, Demolition: Shaped, 15 LB, M2A4.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3, 4 and 5 of this specification, whether they are listed or not.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM-AC, ATTN: FCDD-ACE-QSA , Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-1168 | - | Ammunition Lot Numbering and Ammunition Data Card |
| MIL-STD-1916 | - | DOD Preferred Method for Acceptance of Product |
| MIL-STD-2073-1 | - | Standard Practice for Military Packaging |

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-12560 | - | Armor Plate, Steel, Wrought, Homogeneous (for use in combat-vehicles and for ammunition testing) |

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

U.S. ARMY ARMAMENT CENTER DRAWINGS

| | | |
|---|---|---|
| 8858393 | - | Booster |
| 8858397 | - | Detonator well assembly |
| 8858398 | - | Adaptor, well, detonator |
| 8858399 | - | Well, detonator |
| 8858400 | - | Well, detonator, alternative |
| 8861565 | - | Charge, Demolition: Shaped, 15 LB, M2A4 Loading Assembly |
| 8861567 | - | Sleeve, stand-off |
| 8861568 | - | Closing cap assembly |
| 8861569 | - | Body Assembly |
| 8861570 | - | Container, cone |
| 8861572 | - | Cone assembly |
| 8861579 | - | Cone |
| 8861582 | - | Cap, closing |
| 9230162 | - | Charge, Demolition: Shaped, 15 LB, M2A4, Parts Arrangement |
| 9281170 | - | Packing and Marking for Charge, Demolition: Shaped, 15 LB, M2A4 |
| 13021902 | - | Adapter, priming M1A5 |

(Copies of these drawings are available email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil.)


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US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116/120H – Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallet

(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Default.aspx .)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/1742M - Standard Practice for Radiographic Examination

(Copies of this document are available from https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Material, parts and, assemblies. Materials, parts and assemblies shall comply with all requirements specified in drawings cited in section 2.2.2.

3.2.1 Specific gravity of glass cone. The glass, from which the cone is made, shall be clear potash-soda-lead-silicate having a minimum specific gravity of 3.00.

3.2.2 Specific gravity of booster. The specific gravity of the booster shall be not less than 1.50.

3.2.3 Average explosive weight of loading assemblies. The average explosive weight of the loading assemblies shall be not less than 10.1 lbs (Minimum 10 lbs of Comp B plus 48 grams of booster material).


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3.2.4 Annealing of glass cone. The glass cone shall be fully annealed to relieve strains which might cause failure.

3.2.5 Heat shock test of glass cone. The glass cone shall not break or show cracks when tested for resistance to heat shock.

3.2.6 Radiographic examination. Cavitation shall be permitted only at the top side of the body where the explosive mixture is poured. Cavitation or voids in the loading assembly shall be not greater than ¼ inch depth and 1 inch width in any direction.

3.3 Leakproofness of body assembly. The body assembly, when pressurized to 2.5 pounds per square inch gauge (psig) internally with air, shall not exhibit leakage greater than an amount indicated by a drop of more than 60 percent in 5 seconds.

3.4 Leakproofness of closing cap assembly. The closing cap assembly shall not exhibit leakage when pressurized 2 to 3 psig internally with air for not less than10 seconds.

3.5 Fit of closing cap assembly. The closing cap assembly shall require a force fit to be inserted into the body assembly which does not leave the assembly loose or damaged.

3.6 Fit of stand-off sleeve. The gap between the body and the stand-off sleeve shall not be greater than 0.3 inches when a static load of 20 plus or minus 1/2 pounds is applied.

3.7 Functioning test. The loaded M2A4 shall detonate high order with penetration of a target of Rolled Homogeneous Armor (RHA) in accordance with MIL-DTL-12560, Class 1 or 2, when initiated by an M6 or M7 blasting cap or equivalent military initiation device held in place with the priming adapter. The penetration hole in the plates shall exhibit no evidence of double- or multi-jets and have a minimum penetration of 4 inches. The entrance and exit of the penetration hole diameters of each plate will be two readings at 90 degrees apart for each diameter of the first 4 inches. Any penetration depth and diameter measurements beyond the minimum 4 inch plate thickness requirement will be for performance data collection purposes only.

3.8 Workmanship. All parts and assemblies shall be clean and free of explosive on exterior, chips, cracks, crazes, dirt, grease, rust, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the cleaning agent contaminate them. Exterior surface coatings shall be continuous except for a few light scratches not exposing the base material. All required markings shall be neat and sharply defined. Slight discoloration of the plunger and mold surfaces will be allowed; provide they do not indicate flaws or imperfections which would adversely affect the function of the case.


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## 4. VERIFICATION

TABLE I. Requirements/verification cross reference matrix.

| Method of Verification | | Classes of Verification |
|---|---|---|
| 1- Analysis<br>2- Examination<br>3- Test<br>4- Demonstration | | A - First article<br>B - Conformance |

| Section 3 Requirements | Description | Verification Method | | | | Verification Class | | Section 4 |
|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | A | B | |
| 3.1.1 | First article | X | X | X | | X | | 4.2 |
| 3.1.2 | Conformance inspection | | | X | | | X | 4.3 |
| 3.2 | Material, parts and, assemblies | X | X | X | | X | X | 4.3 - 4.3.3.22 |
| 3.2.1 | Specific gravity of glass cone | X | | | | X | X | 4.4.5 |
| 3.2.2 | Specific gravity of booster | X | | | | X | X | 4.4.8 |
| 3.2.3 | Average weight of loading assemblies | | X | | | X | X | 4.4.9 |
| 3.2.4 | Annealing of glass cone | | X | | | X | X | 4.4.6 |
| 3.2.5 | Heat shock test of glass cone | | | X | | X | X | 4.4.7 |
| 3.2.6 | Radiographic examination | | X | | | X | X | 4.4.10 |
| 3.3 | Leakproofness of body assembly | | X | X | | X | X | 4.4.1 |
| 3.4 | Leakproofness of cap assembly | | X | X | | X | X | 4.4.2 |



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Table I. Requirements/verification cross reference matrix (continued).

| Method of Verification | | Classes of Verification |
|---|---|---|
| 1- Analysis<br>2- Examination<br>3- Test<br>4- 4 – Demonstration | | A - First article<br>B – Conformance |

| Section 3 Requirements | Description | Verification Method | | | | Verification Class | | Section 4 |
|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | A | B | |
| 3.5 | Fit of closing cap assembly | | X | | | X | X | 4.4.3 |
| 3.6 | Fit of stand-off sleeve | | X | X | | X | X | 4.4.4 |
| 3.7 | Functioning test | | | X | | X | X | 4.4.11 |
| 3.8 | Workmanship | | X | | | X | X | 4.3.3.1-4.3.3.22 |
| 5.1 | Packaging and marking | | X | | | X | X | 4.3.3.18 – 4.3.3.22 |
| 5.2 | Unitization | | X | | | | X | 4.3.3.23 |

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)

b. Conformance Inspection (see 4.3)

4.2 First article inspection. A sample shall be subjected to first article inspection in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures and facilities as will be used during regular production.

4.2.1 First article rejection. The first article sample shall be rejected if any of the inspections/tests fail to meet specification or drawing requirements. The Government reserves the right to terminate inspection or test upon failure of any sample to comply with any requirement.

4.2.2 Inspections to be performed. The first article samples shall be subjected to all of the examinations and tests specified in Table II.



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TABLE II. First article inspection.

| Examination or test | Drawing | Sample Size | Requirement paragraph | Inspection method |
|---|---|---|---|---|
| Glass<br>Examination of defects<br>Specific gravity of glass | 9230178 | <br>1<br>10 | <br>3.2.1 | <br>4.3.3.1<br>4.4.5 |
| Well, detonator | 8858399 | 25 | 3.2 | 4.3.3.2 |
| Adapter, well, detonator | 8858398 | 25 | 3.2 | 4.3.3.3 |
| Detonator well assembly | 8858397 | 25 | 3.2 | 4.3.3.4 |
| Well, detonator (alternative) | 8858400 | 25 | 3.2 | 4.3.3.5 |
| Cap, closing | 8861582 | 4.3.2.6 | 3.2 | 4.3.3.6 |
| Closing, cap assembly<br>Examination of defects<br>Leakproofness of closing cap assembly | 8861568 | <br>25 1/<br>100% | <br>3.2<br>3.4 | <br>4.3.3.7<br>4.4.2 |
| Cone<br>Examination of defects<br>Annealing of cone<br>Heat shock | 8861579 | <br>25 1/<br>100%<br>25 | <br>3.2<br>3.2.4<br>3.2.5 | <br>4.3.3.8<br>4.4.6<br>4.4.7 |
| Cone assembly | 8861572 | 25 | 3.2 | 4.3.3.9 |
| Container, cone | 8861570 | 25 | 3.2 | 4.3.3.10 |
| Body assembly, prior to attaching carrying straps<br>Examination of defects<br>Leakproofness<br>Fit of closing cap | 8861569 | <br>25 1/<br>100%<br>25 | <br>3.2<br>3.3<br>3.5 | <br>4.3.3.11<br>4.4.3 |
| Body assembly | 8861569 | 25 | 3.2 | 4.3.3.12 |
| Sleeve, stand-off<br>Examination of defects<br>Fit of stand-off sleeve | 8861567 | <br>25 1/<br>25 | <br>3.2<br>3.6 | <br>4.3.3.13<br>4.4.4 |
| Booster<br>Examination of defects<br>Specific gravity | 8858393 | <br>25 1/<br>10 | <br>3.2<br>3.2.2 | <br>4.3.3.14<br>4.4.8 |
| Body assembly prior to loading | 8861569 | 25 | 3.2 | 4.3.3.15 |
| Loading assembly prior to assembly of closing cap<br>Examination of defects<br>Average weight | 8861565 | <br>25<br>25 | <br>3.2<br>3.2.3 | <br>4.3.3.16<br>4.4.9 |



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TABLE II. First article inspection (continued).

| Examination or test | Drawing | Sample Size | Requirement paragraph | Inspection method |
|---|---|---|---|---|
| Charge, Demolition: Shaped, 15 LB, M2A4 Loading Assembly | 8861565 | | | |
| Examination of defects | | 25 1/ | 3.3.1 | 4.3.3.17 |
| Radiographic examination | | 10 | 3.2.6 | 4.4.10 |
| Function test | | 10 | 3.7 | 4.4.11 |
| Box, fiberboard prior to sealing | 9281170 | 4.3.3.18 | 5.1 | 4.3.3.18 |
| Box, fiberboard | 9281170 | 4.3.3.19 | 5.1 | 4.3.3.19 |
| Sealed packed barrier bag prior to packing in box | 9281170 | | 5.1 | |
| Examination of defects | | 4.3.3.20 | | 4.3.3.20 |
| Heat seal test | | 2/ | | 4.4.12 |
| Packing box prior to securing top | 9281170 | 4.3.3.21 | 5.1 | 4.3.3.21 |
| Packing box | 9281170 | 4.3.3.22 | 5.1 | 4.3.3.22 |
| Unitization | 19-48-4116 19-48-4116/120H | 4.3.3.23 | 5.2 | 4.3.3.23 |

NOTE:
1/ 100% inspection of all drawing characteristics shall be performed.
2/ The test shall be performed at the beginning of each shift on three (3) sealed barrier bags from each sealer.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. Inspection lots shall contain materials from not more than one lot interfix number from one manufacturer (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specifications and revisions. In addition, each lot shall contain materials from not more than one lot interfix number from one manufacturer.

4.3.2 Ammunition lot numbering. Visually verify that a unique ammunition lot number has been assigned to each lot as described in MIL-STD-1168.


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4.3.3 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.3.1 through 4.3.3.22, the definitions of critical, major, and minor defects are provided in paragraph "Definitions" of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).

| 4.3.3.1 Glass. | | | | Drawing number 9230178<br>Next higher assembly 8861579 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Specific gravity of glass | 10 | 3.2.1 | 4.4.5 |
| | | | | |
| Minor | | | | |
| 201 | Evidence of poor workmanship | VL II | 3.8 | Visual |


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| 4.3.3.2 Well, detonator. | | | | Drawing number 8858399<br>Next higher assembly 8858397 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Diameter of cavity | VL IV | 3.2 | Gage |
| 102 | Depth of cavity, minimum (min.) | VL IV | 3.2 | Gage |
| 103 | Outside diameter of body, maximum (max.) | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Diameter of flange, max. | VL II | 3.2 | Gage |
| 202 | Thickness of flange, max. | VL II | 3.2 | Gage |
| 203 | Length of closed end, max. | VL II | 3.2 | Gage |
| 204 | Evidence of poor workmanship | VL II | 3.8 | Visual |


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| 4.3.3.3 Adapter, well, detonator. | | | | Drawing number 8858398<br>Next higher assembly 8858397 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| | | | | |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Pitch diameter of thread, min. | VL IV | 3.2 | Gage |
| 102 | Minor diameter of thread, max. | VL IV | 3.2 | Gage |
| | | | | |
| | | | | |
| Minor | | | | |
| 201 | Overall length, min. | VL II | 3.2 | Gage |
| 202 | Evidence of poor workmanship | VL II | 3.8 | Visual |

| 4.3.3.4 Detonator well assembly. | | | | Drawing number 8858397<br>Next higher assembly 8861568 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Concentricity of inside diameter with thread | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Assembled length of adapter, min. | VL II | 3.2 | Gage |
| 202 | Evidence of poor workmanship | VL II | 3.8 | Visual |

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| 4.3.3.5 Well, detonator (alternative). | | | | Drawing number 8858400<br>Next higher assembly 8861568 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Pitch diameter of thread, min | VL IV | 3.2 | Gage |
| 102 | Minor diameter of thread, max. | VL IV | 3.2 | Gage |
| 103 | Diameter of small cavity, min. | VL IV | 3.2 | Gage |
| 104 | Depth of cavity, min. | VL IV | 3.2 | Gage |
| 105 | Outside diameter of small end, min. | VL IV | 3.2 | Gage |
| 106 | Depth of thread cavity, min. | VL IV | 3.2 | Gage |
| 107 | Concentricity of inside diameter with thread | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Evidence of poor workmanship | VL II | 3.8 | Visual |

| 4.3.3.6 Cap, closing. | | | | Drawing number 8861582<br>Next higher assembly 8861568 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | None defined | | | |
| | | | | |
| Minor | | | | |
| 201 | Internal depth, min. | VL II | 3.2 | Gage |
| 202 | Flange improperly formed | VL II | 3.2 | Visual |
| 203 | Evidence of poor workmanship | VL II | 3.8 | Visual |

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| 4.3.3.7 Closing cap assembly. | | | | Drawing number 8861568<br>Next higher assembly 8861565 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Concentricity of inside diameter of cap with thread | VL IV | 3.2 | Gage |
| 102 | Leakproofness of closing cap assembly | 100% | 3.4 | 4.4.2 |
| | | | | |
| Minor | | | | |
| 201 | Evidence of poor workmanship | VL II | 3.8 | Visual |

| 4.3.3.8 Cone. | | | | Drawing number 8861579<br>Next higher assembly 8861572 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Concentricity of large outside diameter with basic diameters | VL IV | 3.2 | Gage |
| 102 | Location of large basic diameter | VL IV | 3.2 | Gage |
| 103 | Excessive wall thickness variation | VL IV | 3.2 | Gage |
| 104 | Location of small basic diameter | VL IV | 3.2 | Gage |
| 105 | Annealing of cone | 100% | 3.2.6 | 4.4.6 |
| 106 | Heat shock test of cone | 25 | 3.2.7 | 4.4.7 |
| | | | | |
| Minor | | | | |
| 201 | Large outside diameter, max. | VL II | 3.2 | Gage |
| 202 | Total height, max. | VL II | 3.2 | Gage |
| 203 | Evidence of poor workmanship | VL II | 3.8 | Visual |

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| 4.3.3.9 Cone assembly. | | | | Drawing number 8861572<br>Next higher assembly 8861569 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Excessive misalignment between ring, bottom and retainer ring, cone | VL IV | 3.2 | Visual |
| 102 | Cone loose | VL IV | 3.2 | Manual |
| | | | | |
| Minor | | | | |
| 201 | Protective coating missing | VL II | 3.2 | Visual |
| 202 | Evidence of poor workmanship | VL II | 3.8 | Visual |
| | | | | |

| 4.3.3.10 Container, cone. | | | | Drawing number 8861570<br>Next higher assembly 8861569 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Location of upper basic diameter | VL IV | 3.2 | Gage |
| 102 | Concentricity of small inside diameter with basic diameter | VL IV | 3.2 | Gage |
| 103 | Large inside diameter | VL IV | 3.2 | Gage |
| 104 | Location of lower basic diameter | VL IV | 3.2 | Gage |
| | | | | |
| Minor | | | | |
| 201 | Evidence of poor workmanship | VL II | 3.8 | Visual |



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| 4.3.3.11 Body assembly, prior to attaching carrying straps. | | | | Drawing number 8861569 |
|---|---|---|---|---|
| | | | | Next higher assembly 8861565 |
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Concentricity of container of cone | VL IV | 3.2 | Gage |
| 102 | Cone assembly insecure | VL IV | 3.2 | Manual |
| 103 | Reinforcing ring missing or improperly crimped | VL IV | 3.2 | Visual |
| 104 | Bare spot on interior surface | VL IV | 3.2 | Visual |
| 105 | Leakproofness of body assembly | 100% | 3.3 | 4.4.1 |
| 106 | Fit of closing cap assembly | 25 | 3.5 | 4.4.3 |
| | | | | |
| Minor | | | | |
| 201 | Gap at base, max. | VL II | 3.2 | Gage |
| 202 | Evidence of poor workmanship | VL II | 3.8 | Visual |

| 4.3.3.12 Body assembly. | | | | Drawing number 8861569 |
|---|---|---|---|---|
| | | | | Next higher assembly 8861565 |
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | None defined | | | |
| | | | | |
| Minor | | | | |
| 201 | Carrying strap missing or improperly assembled | VL II | 3.2 | Visual/Manual |
| 202 | Evidence of poor workmanship | VL II | 3.8 | Visual |



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| 4.3.3.13 Sleeve, stand-off. | | | | Drawing number 8861567<br>Next higher assembly 9230162 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Total length, max. | VL IV | 3.2 | Gage |
| 102 | Distance shoulder to end | VL IV | 3.2 | Gage |
| 103 | Fit of stand-off sleeve | 25 | 3.6 | 4.4.4 |
| | | | | |
| Minor | | | | |
| 202 | Evidence of poor workmanship | VL II | 3.8 | Visual |

| 4.3.3.14 Booster. | | | | Drawing number 8858393<br>Next higher assembly 8861565 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Concentricity of small cavity to outside diameter | VL IV | 3.2 | Gage |
| 102 | Diameter of small cavity, min. | VL IV | 3.2 | Gage |
| 103 | Depth of Large cavity, min. | VL IV | 3.2 | Gage |
| 104 | Specific gravity | 4.4.8 | 3.2.4 | 4.4.8 |
| | | | | |
| Minor | | | | |
| 201 | Depth of small cavity, min. | VL II | 3.2 | Gage |
| 202 | Diameter of large cavity, min. | VL II | 3.2 | Gage |
| 203 | Total length, max. | VL II | 3.2 | Gage |
| 204 | Radius missing | VL II | 3.2 | Visual |
| 205 | Evidence of poor workmanship | VL II | 3.8 | Visual |



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| 4.3.3.15 Body assembly prior to loading. | | | | Drawing number 8861569 |
|---|---|---|---|---|
| | | | | Next higher assembly 8861565 |
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| | | | | |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Outside diameter, max. | VL IV | 3.2 | Gage |
| 102 | Interior protective finish discontinuous, scratched or contains bare spots | VL IV 1/ | 3.2 | Visual |
| 103 | Foreign matter present in interior | VL IV | 3.2 | Visual |
| 104 | Carrying strap damaged | VL IV | 3.2 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Evidence of poor workmanship | VL II | 3.8 | Visual |
| NOTE:<br>1/ Any scratch deep enough to show the base material of the item | | | | |

| 4.3.3.16 Loading assembly prior to assembly of closing cap. | | | | Drawing number 8861565 |
|---|---|---|---|---|
| | | | | Next higher assembly 9281170 |
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Component missing, loose or improperly assembled | VL IV | 3.2 | Visual/Manual |
| 102 | Body assembly damaged | VL IV | 3.2 | Visual |
| 103 | Average weight | 25 | 3.2.5 | 4.4.9 |
| | | | | |
| Minor | | | | |
| 201 | Depth to charge, min. | VL II | 3.2 | Gage |
| 202 | Evidence of poor workmanship | VL II | 3.8 | Visual |



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| 4.3.3.17 Charge, Demolition: Shaped, 15 LB, M2A4 Loading Assembly. | | | | Drawing number 8861565<br>Next higher assembly 9281170 |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Gage does not enter detonator well freely to full length | VL IV | 3.2 | Gage |
| 102 | Closing cap assembly loose | VL IV | 3.2 | Manual |
| 103 | Protective tape missing | VL IV | 3.2 | Visual |
| 104 | Radiographic examination | 25 | 3.2.6 | 4.4.10 |
| 105 | Functioning test | 10 1/ | 3.7 | 4.4.11 |
| | | | | |
| Minor | | | | |
| 201 | Total length | VL II | 3.2 | Gage |
| 202 | Marking missing, incorrect or illegible | VL II | 3.2 | Visual |
| 203 | Evidence of poor workmanship | VL II | 3.8 | Visual |
| | | | | |

NOTE:
1/ After three (3) consecutive lots have met the criteria, a sample of five (5) test units shall be selected and function tested out of each lot produced. If one or more loaded units fail to function (or a new First Article is required), the sampling shall return to the original test plan and continue until three (3) consecutive lots have complied with the acceptance criteria specified.



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| 4.3.3.18 Box, fiberboard prior to sealing. | | | | Drawing number 9281170<br>Next higher assembly 19-48-4116 19-48-4116/120H |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Component missing or improperly packed | VL IV | 5.1 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Presence of M1A5 adapters (Dwg #13021902) | VL II | 5.1 | Visual |
| 202 | Evidence of poor workmanship | VL II | 5.1 | Visual |



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| 4.3.3.19 Box, fiberboard. | | | | Drawing number 9281170<br>Next higher assembly 19-48-4116 19-48-4116/120H |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Sealing strip incomplete or loose | VL IV | 5.1 | Visual |
| 102 | Box punctured, torn or cut | VL IV | 5.1 | Visual |
| 103 | Corner not blunted | VL IV | 5.1 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Marking missing, incorrect or illegible | VL II | 5.1 | Visual |
| 202 | Contents loose | VL II | 5.1 | Manual |
| 203 | Evidence of poor workmanship | VL II | 5.1 | Visual |



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| 4.3.3.20 Sealed packed barrier bag prior to packing in box. | | | | Drawing number 9281170<br>Next higher assembly 19-48-4116 19-48-4116/120H |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal | VL IV | 5.1 | Visual |
| 102 | Evidence of crushed box (see 1/) | VL IV | 5.1 | Visual |
| 103 | All excess air not exhausted from bag prior to sealing | VL IV | 5.1 | Visual |
| 104 | Heat seal test | 2/ | 5.1 | 4.4.12 |
| | | | | |
| Minor | | | | |
| 201 | Heat seal of barrier bag located so that one or more subsequent seals cannot be applied | VL II | 5.1 | Visual |
| 202 | Marking of bag improper or unidentifiable | VL II | 5.1 | Visual |
| 203 | Evidence of poor workmanship | VL II | 5.1 | Visual |
| | | | | |

NOTE:
1/ Evidence of visible dents or deformations, torn or punctured surfaces and or collapsed regions on the box
2/ The test shall be performed at the beginning of each shift on one (1) sealed barrier bag from each sealer. For sampling purposes, consider a shift to be no be more than ten (10) hours



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| 4.3.3.21 Packing box prior to securing top. | | | | Drawing number 9281170<br>Next higher assembly 19-48-4116 19-48-4116/120H |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Number of sealed bags incorrect | VL IV | 5.1 | Visual |
| 102 | Any packing component missing | VL IV | 5.1 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Insufficient filler material | VL II | 5.1 | Visual |
| 202 | Evidence of poor workmanship | VL II | 5.1 | Visual |



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| 4.3.3.22 Packing box. | | | | Drawing number 9281170<br>Next higher assembly 19-48-4116 19-48-4116/120H |
|---|---|---|---|---|
| | Examination or test | Conformance criteria | Requirement paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Board broken or split | VL IV | 5.1 | Visual |
| 102 | Contents exposed or loose | VL IV | 5.1 | Visual/Manual |
| 103 | Hardware or strapping missing, loose or broken | VL IV | 5.1 | Visual/Manual |
| 104 | NSN number missing or misleading | VL IV | 5.1 | Visual |
| | | | | |
| Minor | | | | |
| 201 | Marking missing, incorrect or illegible | VL II | 5.1 | Visual |
| 202 | Evidence of poor workmanship | VL II | 5.1 | Visual |
| 203 | Metallic seal missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly | VL II | 5.1 | Visual/Manual |


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| 4.3.3.23 Unitization. | | | Drawing Number 19-48-4116 19-48-4116/120H<br>Next Higher Assembly 13025692 | |
|---|---|---|---|---|
| | Examination | Conformance Criteria | Requirement Paragraph | Inspection method |
| Critical | None defined | | | |
| | | | | |
| Major | | | | |
| 101 | Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location | VL II | 5.2 | Visual |
| 102 | Pallet load markings -including 2D bar codes - missing, incorrect, illegible, or not in proper location | VL II | 5.2 | Visual |
| 103 | Filler assembly not built, treated, or marked correctly | VL II | 5.2 | Visual |
| 104 | Boxes incorrectly oriented or positioned on pallet | VL II | 5.2 | Visual |
| 105 | Strapping not applied correctly or incorrect type | VL II | 5.2 | Visual |
| 106 | Unit load not tightly strapped or any strap loose | VL II | 5.2 | Visual / Manual |
| | | | | |
| Minor | | | | |
| 201 | Workmanship | VL I | 5.1 | Visual |

4.4 Test methods and procedures.

4.4.1 Leakproofness of body assembly. The unit shall be assembled to the test equipment, the required internal pressure applied, and observation made for leakage for the specified period. Any assembly exhibiting more than the specified leakage shall be classified defective and removed from the lot.

4.4.2 Leakproofness of closing cap assembly. The unit shall be assembled to the test equipment, the required internal pressure applied, and observation made for leakage for the specified period. Any assembly exhibiting leakage shall be classified defective and removed from the lot.

4.4.3 Fit of closing cap assembly. Unless otherwise noted, a sample of twenty-five (25) closing cap assemblies and twenty-five (25) body assemblies minus cone assembly shall be selected from each lot for this test. The closing cap assembly shall be assembled to its cone container using applicable equipment. Failure to require a force fit that leaves the assembly


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loose or damaged shall class the full assembly defective. Failure of one or more full assemblies in meeting the requirement shall reject both lots of components.

4.4.4 Fit of stand-off sleeve. Unless otherwise noted, a sample of twenty-five (25) stand-off sleeves and twenty-five (25) body assemblies shall be selected during first article and from each lot for this test. The samples of stand-off sleeves and body assemblies shall be segregated into random combinations of one unit of each. The stand-off sleeve shall be placed on a level surface, the body assembly placed thereon, and the required load applied to the top of the body assembly. The gap between the components shall be measured, the assembly picked up and given a sharp shake toward ejection of the sleeve. If the gap is excessive or the sleeve disassembles when shaken, the assembly shall be classed defective. Failure of one or more combinations in meeting either requirement shall reject both lots of components.

4.4.5 Specific gravity of glass cone. Ten (10) sample cones in each lot shall be checked to determine compliance with the density requirements. The density shall be obtained by weighing the sample cone, first in air and then, while freely suspended in distilled water. The temperature of the distilled water shall be 68 degrees Fahrenheit (ºF) +/- 2 ºF (see TABLE III). The weight of the cone should be to the nearest .01 grams. The density shall be calculated by means of the following formula:

$$\textit{Specific gravity} = \frac{\textit{Weight of glass cone dry}}{\textit{(Weight of glass cone dry minus Weight of glass cone immersed in water)}} \times \textit{Specific gravity of } H_2O \textit{ at test temperature (see Table III)}$$

If any sample cone fails to meet this requirement, this failure shall be considered cause for rejection of the lot represented.

4.4.6 Annealing of glass cone. The cones shall be examined by means of a polariscope to determine whether the glass has been relieved of all strains. Cones showing evidence of unrelieved strains shall be rejected. Any glass cone which fails to comply with the applicable requirements shall be classed defective, and this sample shall be rejected.

4.4.7 Heat shock test of glass cone. The sample cone shall be conditioned at 45 degrees Fahrenheit (ºF) +/- 36 degrees ºF for twenty (20) minutes minimum. The convex surface of the cones shall then be placed for two minutes, minimum, into water maintained at 203 degrees ºF +/- 41 degrees ºF. Any sample which breaks or shows cracks shall reject the sample and lot of cones.

4.4.8 Specific gravity of booster. (See Drawing (dwg.) 8858393) A sample of ten (10) boosters shall be selected from each lot for this test. If any booster fails to meet the requirement specified, the lot shall be rejected. The booster shall be weighed on a balance and the weight recorded as dry weight. The booster shall then be weighed while completely submerged in water, taking care to remove all air bubbles remaining on the surfaces of the booster. The temperature of the distilled water shall be 68 degrees ºF +/- 2 ºF. This weight shall be recorded as weight



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under water. (When submerged, the position of the booster shall be such that no air is entrapped.)The weight of the booster should be to the nearest .01 grams. Specific gravity shall be calculated as follows:

$$\textit{Specific gravity} = \frac{\textit{Weight of booster dry}}{\textit{(Weight of booster dry minus Weight of booster immersed in water)}} \times \textit{Specific gravity of } H_2O \textit{ at test temperature (see Table III)}$$

Table III. Specific gravity of water in test temperature range.

| Temp °F | Temp °C | Specific Gravity of $H_2O$ |
|---|---|---|
| 66 | 18.89 | 0.9984 |
| 67 | 19.44 | 0.9983 |
| 68 | 20.00 | 0.9982 |
| 69 | 20.56 | 0.9981 |
| 70 | 21.11 | 0.9980 |

4.4.9 Weight of charge and average weight. The loading assembly shall be weighed on a scale capable of determining weights within an accuracy of two (2) ounces. The mean average weight shall be obtained from twenty-five (25) parts assemblies (dwg. 8861569) and subtracted from the weight of the loading assembly (dwg. 8861565) to determine compliance with the requirement. If the mean average weight of the sample assemblies fails to meet the weight requirement, all samples shall be classified as defective and the lot shall be rejected.

4.4.10 Radiographic examination. Radiographic equipment, operations and procedures shall be in accordance with ASTM E1742/E1742M. The radiographic examination shall detect voids, cracks, cavitation, separation, and cracking of the explosive and presence of any foreign material. The assembly shall be radiographically inspected two (2) times. The assembly shall be inspected along its length, rotated approximately ninety degrees (90°) on its cylindrical axis, and inspected a second time. Any assembly which does not meet the criteria shall reject the sample(s).

4.4.11 Functioning test. Ten (10) test units shall be selected from each lot. The test shall be detonated using an M6 or M7 blasting cap (see 6.2j) or equivalent military initiation device held in place with the priming adapter. The test shall also be conducted at local temperatures and temperatures shall be recorded for data collection purposes only. Any test unit which fails to detonate high order shall be classified as defective and the lot is to be rejected. Observation shall be made for the presence of undetonated charge and for evidence of double- or multi-jets. Presence of either shall classify a unit defective. The RHA target shall be of a size not less than twelve (12) inches in total thickness with a maximum of six (6) plates making up that thickness. The first four (4) inches shall be made up of a maximum of two (2) plates at two (2) inches minimum thickness each. The surface area shall not be less than 144 inches made up with one (1) plate. Any penetration depth not meeting the requirements shall be classified as a unit failure and



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the lot is to be rejected. After three (3) consecutive lots have met the criteria, a sample of five (5) test units shall be selected and function tested out of each lot produced. If one or more loaded units fail to function, the lot shall be rejected and the sampling shall return to the original test plan.

4.4.12 Heat seal test of sealed bag (destructive). Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-STD-2073-1, para 5.2.3.6.1 Method 41 Water vaporproof bag, heat sealed. Samples that do not meet the heat seal requirement shall be rejected and all samples tested to last successful test.

## 5. PACKAGING

5.1 Packaging and marking. The materials, components, and assemblies shall comply with Drawing 9281170 for the M2A4 and all their subsidiary drawings.

5.2 Unitization. Unitize M2A4 in accordance with Drawing 19-48-4116/120H and their subsidiary drawings.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The 15 lb. Shaped Demolition Charge is primarily employed for burring holes in earth, metal, concrete, and paved or unpaved roads

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Requirement for acceptance inspection equipment (AIE) designs.

d. Requirement for Alternate Acceptance Plans.

e. Requirements for Ammunition Lot Numbering, if applicable.

f. Requirement for propellant and energetic materials IAW MIL-STD-1171

g. Certificate of Conformance for each lot or shipment of product.

h. Preservation and packaging. Preservation, packaging, packing and marking should be in accordance with Dwg. 9281170.



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i. Radiographic personnel should be qualified in accordance with NAS 410.

j. The 15 lb. Shaped Demolition Charge should be functioned with either an M6 Blasting Cap in accordance with MIL-PRF-32293A or an M7 Blasting Cap in accordance with MIL-DTL-45469F and Drawing 8830948.

6.3 Radiographic examination. Complete results of the radiographic examination should be available to the Government through the Contracting Officer. Distribution should include: DEVCOM-AC, ATTN: FCDD-ACE-QEA & FCDD-ACE-QWE, Picatinny, NJ 07806-5000.

6.4 Acceptance inspection equipment. Complete results of the acceptance inspection equipment should be available to the Government through the Contracting Officer. Distribution should include: DEVCOM-AC, ATTN: FCDD-ACE-QWE , Picatinny, NJ 07806-5000.

6.4 Subject term (key word) listing.

Explosive
Composition B

6.5 Superseding explanatory note. This revision combines MIL-DTL-13139F with MIL-DTL-2156D into a single specification. MIL-DTL-1319F covered the parts for manufacturing the CHARGE, DEMOLITION, SHAPED, 15LB.-M2A4, and MIL-DTL-2156D covered the loading, assembling and packing for the same item. As a result of the combination of the two specifications, MIL-DTL-13139F has been cancelled.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army-AR
Navy -OS
Air Force - 70

Preparing Activity:
Army-AR
(Project: 1375-2024-014)

Review Activity
Navy – AS, NP
Air Force – 11

Civil agency:
GSA-FAS

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